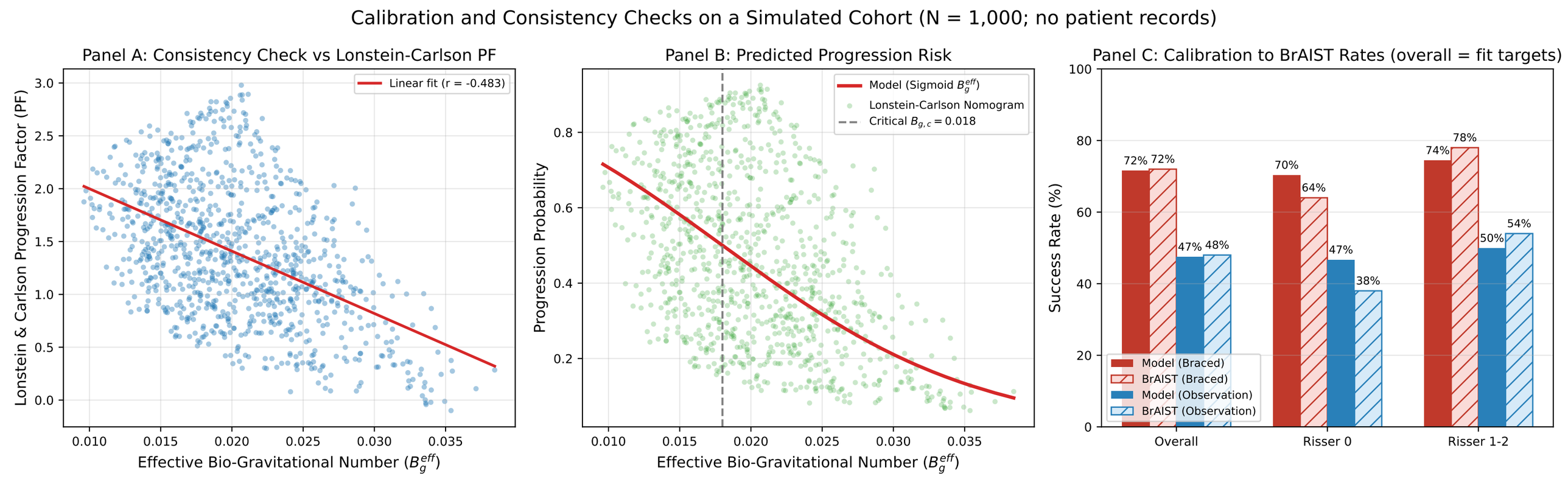

Calibration and Consistency Checks on a Simulated Cohort (N = 1,000; no patient records)
Panel A: Consistency Check vs Lonstein-Carlson PF
Linear fit (r = -0.483)
Lonstein & Carlson Progression Factor (PF)
Effective Bio-Gravitational Number (B_g^{eff})
Panel B: Predicted Progression Risk
Model (Sigmoid B_g^{eff})
Lonstein-Carlson Nomogram
Critical $B_{g,c} = 0.018$
Progression Probability
Effective Bio-Gravitational Number (B_g^{eff})
Panel C: Calibration to BrAIST Rates (overall = fit targets)
Success Rate (%)
72%
72%
47%
48%
70%
64%
47%
38%
74%
78%
50%
54%
Model (Braced)
BrAIST (Braced)
Model (Observation)
BrAIST (Observation)
Overall
Risser 0
Risser 1-2